The Golden Pi White Paper

$$\hat{\pi} = \frac{4}{\sqrt{\varphi}} = 3.144605511\ldots$$

A Unified Construction of the Circle Constant Through the Golden Ratio, the Great Pyramid, and the Eye of Horus

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Abstract. This white paper consolidates the full body of research on the Golden Pi constant $\hat{\pi} = 4/\sqrt{\varphi} = 3.144605511\ldots$, a constructible, algebraic circle constant derived entirely from the golden ratio φ. We present its algebraic origin, the *instrumentum identity* that produces it in closed form, its encoding in the Great Pyramid of Giza through the Royal Cubit and the Seked slope, the $abc = 64$ golden triangle that reproduces the pyramid face angle, and its relation to the Eye of Horus fractions. Every numerical claim is exact or explicitly bounded.

1. Introduction

The ratio of a circle's circumference to its diameter is conventionally taken to be $\pi \approx 3.14159265$, a *transcendental* number whose decimal expansion is neither terminating nor eventually periodic. Since Lindemann's 1882 proof that π is transcendental, it is known that π cannot be constructed with a compass and straightedge, and the classical problem of *squaring the circle* is thereby impossible in the Euclidean plane.

This paper examines an alternative circle constant derived from the golden ratio:

$$\boxed{\hat{\pi} \;=\; \frac{4}{\sqrt{\varphi}} \;=\; 3.144605511029693\ldots} \tag{1}$$

where $\varphi = (1+\sqrt{5})/2 \approx 1.61803398875$ is the golden ratio. Unlike the conventional π, the constant $\hat{\pi}$ is *algebraic*: it is built from a single square root of the golden ratio and is therefore constructible. Its deviation from the transcendental π is small:

$$\frac{\hat{\pi} - \pi}{\pi} = 9.59 \times 10^{-4} \quad (\approx 0.096\%). \tag{2}$$

This paper does not claim that $\hat{\pi}$ replaces the standard π in analytic mathematics. Rather, it documents the observation that the number $4/\sqrt{\varphi}$ arises with striking regularity across independent geometric, architectural, and symbolic systems, and that a family of closed-form identities reproduces it exactly.

2. The Golden Ratio and Its Algebraic Field

The golden ratio is the unique positive solution of $\varphi^2 = \varphi + 1$:

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad \frac{1}{\varphi} = \varphi - 1, \qquad \varphi^2 = \varphi + 1. \tag{3}$$

Its square root appears throughout this work:

$$\sqrt{\varphi} = \sqrt{\frac{1+\sqrt{5}}{2}} = 1.27201964951\ldots \tag{4}$$

The number $\sqrt{\varphi}$ has a remarkable double interpretation that recurs throughout this paper: it is simultaneously

- an algebraic square root (degree 2, hence *constructible*), and
- a close rational approximation to the Egyptian *Seked* ratio 7/5.5, the rise-over-run slope of the Great Pyramid's face.

Indeed, $7/5.5 = 1.272727\ldots$ and $\sqrt{\varphi} = 1.272019\ldots$ agree to within 0.056%. This single coincidence — the golden ratio's square root approximating the pyramid's slope — is the thread that unifies every result that follows.

3. The Instrumentum Identity

The Golden Pi constant emerges from a closed-form identity of two variables. Define

$$f(x,y) = \frac{4\,x\,y^2}{(y^2+x^2)\,\sqrt{y^2-x^2}}, \qquad y > x > 0. \tag{5}$$

Claim. When the ratio of the two variables equals the square root of the golden ratio, $y/x = \sqrt{\varphi}$, the identity returns exactly $4/\sqrt{\varphi}$:

$$f(x,\, x\sqrt{\varphi}) = \frac{4}{\sqrt{\varphi}} = \hat{\pi}. \tag{6}$$

Proof. Set $x = 1$, $y = \sqrt{\varphi}$. Substituting into (5):

$$f = \frac{4\cdot\sqrt{\varphi}}{(\varphi+1)\,\sqrt{\varphi-1}} = \frac{4\sqrt{\varphi}}{\varphi^2\,\sqrt{\varphi-1}}.$$

Using $\varphi^2 = \varphi + 1$ and $1/\varphi = \varphi - 1$, this simplifies exactly to $4/\sqrt{\varphi}$. Numerically,

$$f(1, \sqrt{\varphi}) = 3.144605511029693\ldots = \hat{\pi}.$$

The identity is *exact* — there is no approximation anywhere in the derivation. The only input is the golden ratio; the output is the circle constant $\hat{\pi}$.

4. The Great Pyramid of Giza

The Great Pyramid encodes the same golden structure in stone. Its canonical dimensions, in the standard model, are a height of 280 royal cubits and a base side of 440 royal cubits.

4.1. The Royal Cubit and the Golden Ratio

If we define the Royal Cubit in terms of the golden ratio as

$$\mathrm{RC} \;=\; \frac{\varphi^2}{5} \;=\; 0.52360679775\ \mathrm{m}, \tag{7}$$

then the pyramid's dimensions become exact golden multiples:

$$\text{height} = 280\,\mathrm{RC} = 56\varphi^2 = 146.609\ \mathrm{m}, \qquad \text{base} = 440\,\mathrm{RC} = 88\varphi^2 = 230.387\ \mathrm{m}. \tag{8}$$

The ratio of height to base, $56 : 88 = 7 : 11$, mirrors the Seked slope directly.

4.2. The Cubit Bridge: $\hat{\pi} \approx 6\,\mathrm{RC}$

A striking coincidence links the circle constant, the cubit, and the number six. The conventional transcendental π divided by the golden cubit is almost exactly six:

$$\frac{\pi}{\mathrm{RC}} = 5.999908\ldots \;\approx\; 6, \qquad \frac{\hat{\pi}}{\mathrm{RC}} = 6.005662\ldots \;\approx\; 6. \tag{9}$$

Equivalently,

$$6\,\mathrm{RC} = 6 \cdot \frac{\varphi^2}{5} = 3.141641\ldots \;\approx\; \hat{\pi} \;=\; 3.144606. \tag{10}$$

Thus the golden-ratio cubit "bridges" the two circle constants to the integer six, connecting an architectural unit to a mathematical constant.

4.3. The Seked Slope

The Seked is the Egyptian measure of slope: the horizontal run, in palms, per cubit of rise. The Great Pyramid's face uses a Seked of 5.5 palms, i.e. a rise-over-run of

$$\text{seked} = \frac{7}{5.5} = \frac{14}{11} = 1.272727\ldots \qquad \Longleftrightarrow \qquad \text{face angle} = \arctan\frac{14}{11} = 51.8428°. \tag{11}$$

The Seked ratio is an excellent rational approximation to $\sqrt{\varphi}$:

$$\frac{7}{5.5} = 1.272727\ldots, \qquad \sqrt{\varphi} = 1.272019\ldots, \qquad \text{relative error} \approx 0.056\%. \tag{12}$$

When the Seked inverse $7/5.5$ is placed in the square-root slot of the Golden Pi expression, it returns the classical Egyptian rational approximation to π:

$$\frac{4}{7/5.5} = \frac{22}{7} = 3.142857\ldots, \qquad \text{vs.} \qquad \frac{4}{\sqrt{\varphi}} = \hat{\pi} = 3.144606. \tag{13}$$

The Seked therefore sits exactly at the convergence of three numbers: $\sqrt{\varphi}$ (the golden construction), $22/7$ (the classical π), and $\hat{\pi}$ (Golden Pi).

5. The Golden Pi Triangle: $abc = 64$

A single right triangle unifies the golden ratio, the pyramid angle, and the Eye of Horus. Define three positive numbers:

$$a = \hat{\pi} = \frac{4}{\sqrt{\varphi}} \qquad \text{(Golden Pi)}, \tag{14}$$

$$b = 4, \tag{15}$$

$$c = \sqrt{a^2 + b^2} \qquad \text{(hypotenuse of a right triangle with legs } a, b\text{)}. \tag{16}$$

Claim. The hypotenuse is $c = 4\sqrt{\varphi}$, and the product of all three sides is exactly 64:

$$abc = \left(\frac{4}{\sqrt{\varphi}}\right)(4)(4\sqrt{\varphi}) = 64. \tag{17}$$

Proof. Since $a^2 = 16/\varphi$ and $b^2 = 16$,

$$c^2 = a^2 + b^2 = \frac{16}{\varphi} + 16 = 16\left(1 + \frac{1}{\varphi}\right) = 16\varphi,$$

using $1 + 1/\varphi = \varphi$. Therefore $c = \sqrt{16\varphi} = 4\sqrt{\varphi}$. Multiplying, the $\sqrt{\varphi}$ in the denominator of a cancels the $\sqrt{\varphi}$ in c:

$$abc = \frac{4}{\sqrt{\varphi}} \cdot 4 \cdot 4\sqrt{\varphi} = 64.$$

The cancellation is exact. There is no error term. The irrational parts annihilate.

5.1. Golden Geometric Progression

The three sides form a *geometric progression* with common ratio $\sqrt{\varphi}$:

$$b/a = \frac{4}{4/\sqrt{\varphi}} = \sqrt{\varphi}, \qquad c/b = \frac{4\sqrt{\varphi}}{4} = \sqrt{\varphi}.$$

Hence the triple is $\{a,\ a\sqrt{\varphi},\ a\varphi\} = \{3.1446,\ 4.0000,\ 5.0881\}$, where $c = a\varphi = \hat{\pi} \cdot \varphi$.

5.2. The Emergent Pyramid Angle

The acute angles of this right triangle are determined by the ratio of its legs. The complement angle is

$$90^\circ - \arctan\left(\frac{a}{b}\right) = 90^\circ - \arctan\left(\frac{1}{\sqrt{\varphi}}\right) = 51.8273^\circ. \tag{18}$$

This reproduces the Great Pyramid's face angle to within one arcminute:

Quantity	**Value**	**Source**
Golden Pi triangle complement	51.8273°	$\arctan(1/\sqrt{\varphi})$
Great Pyramid face angle	51.8428°	Seked $\arctan(14/11)$
Difference	0.0155°	≈ 0.9 arcmin

The same triangle that resolves to $abc = 64$ also encodes the pyramid's slope.

5.3. The Eye of Horus Completion

The Eye of Horus is a sequence of six unit fractions, each a successive power of one half:

$$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}+\frac{1}{64}=\frac{63}{64}. \tag{19}$$

The sum famously stops at 63/64, leaving 1/64 missing. The Golden Pi triangle produces $64 = 2^6$ exactly:

System	Result	Base	Status
Eye of Horus	63/64	binary $(1/2)^n$	missing 1/64
Golden Pi triangle	64	golden φ	**exact**

Where the Eye of Horus reaches 63 and requires the divine 1/64, the Golden Pi triangle reaches 64 algebraically — the irrational parts annihilate and supply the very missing piece.

6. The Golden Pi Expression

The constant $\hat{\pi}$ can also be written through a self-referential expression involving φ and the angle measures 360°:

$$\frac{\dfrac{360}{90\cdot\sqrt{\varphi}\cdot\varphi^2}}{\dfrac{\varphi^2}{2}} = \frac{360}{90\sqrt{\varphi}} = \frac{4}{\sqrt{\varphi}} = \hat{\pi}. \tag{20}$$

The factors of φ^2 cancel, and $360/90 = 4$, collapsing to the same $4/\sqrt{\varphi}$. This form makes explicit that $\hat{\pi}$ is scale-free in the golden ratio — the quadratic factors are scaffolding that cancels identically.

7. The Seked Convergence: Height $\times$ Golden Pi $=$ 2 Base

A striking symmetry emerges when Golden Pi is combined with the pyramid's canonical dimensions ($h = 280$ cubits, base $= 440$ cubits). Compute the product of the height and Golden Pi, and compare to twice the base:

$$h\cdot\hat{\pi} = 280\cdot\frac{4}{\sqrt{\varphi}} = 880.4895\ldots \qquad 2\,\text{base} = 880. \tag{21}$$

$$\frac{h\cdot\hat{\pi}}{2\,\text{base}} = 1.0005563\ldots \qquad \Longleftrightarrow \qquad \text{error} = +0.056\%. \tag{22}$$

This is *exactly* the same magnitude of error as the relation between the Seked slope and $\sqrt{\varphi}$, and between $\hat{\pi}$ and 22/7. The equality $h\cdot\hat{\pi} = 2\,\text{base}$ would be *exact* if and only if $\sqrt{\varphi} = 14/11$:

$$280\cdot\frac{4}{\sqrt{\varphi}} = 880 \quad \Longleftrightarrow \quad \sqrt{\varphi} = \frac{1120}{880} = \frac{14}{11} = 1.272727\ldots \tag{23}$$

But 14/11 is precisely the Seked slope. The pyramid therefore encodes both circle constants to the *same* 0.056% tolerance, from a single cause:

Relation	Value	Error
Perimeter/height $= 2(22/7)$	6.285714	0.0000% (exact)
$h \cdot \hat{\pi} = 2\,\text{base}$	1.000556	+0.0556%
$\sqrt{\varphi}$ vs Seked 14/11	1.272020 vs 1.272727	−0.0556%
$\hat{\pi}$ vs 22/7	3.144606 vs 3.142857	+0.0556%

7.1. The Unified Interpretation

The Seked is the wedge that makes the pyramid *simultaneously* an encoding of the classical $\pi = 22/7$ (via the perimeter-to-height ratio) and of Golden Pi $\hat{\pi} = 4/\sqrt{\varphi}$ (via the height-times-$\hat{\pi}$ equals twice-base relation).

The reason both work is that the pyramid's slope ratio is, to within 0.056%, the golden ratio's square root:

$$\underbrace{\frac{14}{11}}_{\text{Seked slope}} \approx \underbrace{\sqrt{\varphi}}_{\text{golden square root}} \qquad (0.056\%). \tag{24}$$

Because $\sqrt{\varphi}$ sits at the convergence of the Seked (an architectural choice) and the golden construction (an algebraic fact), any circle constant that passes through $4/\sqrt{\varphi}$ — Golden Pi — will "fit" the pyramid's stones with the same precision as the historical 22/7. The pyramid is thus not a proof of either constant, but a shared analog: its slope is the golden square root, and both π and $\hat{\pi}$ are its natural shadows.

8. The Nested-Radical Closed Form

Golden Pi admits a pure square-root form involving no division by the golden ratio, only nested radicals of five:

$$\hat{\pi} = \frac{4}{\sqrt{\varphi}} = \sqrt{\sqrt{320} - 8} = \sqrt{8\sqrt{5} - 8} = \sqrt{8(\sqrt{5} - 1)} = 3.144605511029693\ldots \tag{25}$$

The equivalence follows from the reciprocal golden-ratio relation $\sqrt{5} - 1 = 2/\varphi$:

$$\sqrt{8(\sqrt{5} - 1)} = \sqrt{\frac{16}{\varphi}} = \frac{4}{\sqrt{\varphi}} = \hat{\pi}. \tag{26}$$

This form is notable because it encodes Golden Pi using only the constant $\sqrt{5}$ (the seed of the golden ratio) under nested square roots — no fractions, no transcendental functions. It is a compact, exact, purely radical construction, and it was discovered as the slider maximum in a Desmos model of the instrumentum identity.

8.1. The AREAS Desmos Construction

A self-consistent Desmos model ("AREAS") renders the same identity across three representations, each collapsing to $\hat{\pi}$:

1. $\pi_c = 4/\sqrt{\varphi}$ — the defining closed form (slider max set to $\sqrt{\sqrt{320} - 8}$, the nested-radical form above).

2. $s = \dfrac{2\pi_c\, r}{8}$ — a derived quantity. At $r = 4$, this collapses to $s = \pi_c$ exactly (since $2r/8 = 1$).

3. $\pi_t = \dfrac{4\, s\, r^2}{(r^2 + s^2)\sqrt{r^2 - s^2}}$ — the instrumentum identity fed with $(a = s,\ b = r)$, returning $\hat{\pi}$.

4. $\pi_p = \dfrac{4\, A_c^2 A_s}{(A_c^2 + A_s^2)\sqrt{A_c^2 - A_s^2}}$ — the same identity relabeled on areas A_c, A_s, returning $\hat{\pi}$.

Every expression evaluates to $3.14460551103 = \hat{\pi}$ exactly. The construction is a faithful demonstration of Golden Pi's self-consistency: $\hat{\pi}$ reproduces itself through the identity across the closed form, a derived quantity, and area-labeled variables.

9. Numerical Verification Summary

All constants used in this paper, to twelve decimal places:

Quantity	**Symbol**	**Value**
Golden ratio	φ	1.618033988750
Square root of φ	$\sqrt{\varphi}$	1.272019649514
Golden Pi	$\hat{\pi} = 4/\sqrt{\varphi}$	3.144605511030
Nested-radical form	$\sqrt{\sqrt{320} - 8} = \sqrt{8(\sqrt{5} - 1)}$	3.144605511030
Conventional pi	π	3.141592653590
Relative deviation	$(\hat{\pi} - \pi)/\pi$	$+9.59 \times 10^{-4}$
Royal Cubit	$\varphi^2/5$	0.523606797750 m
Seked slope	$7/5.5 = 14/11$	1.272727272727
Seked face angle	$\arctan(14/11)$	51.84277°
Instrumentum identity	$f(1, \sqrt{\varphi})$	3.144605511030
Golden triangle product	abc	64.0000000000
Golden triangle complement	$90^\circ - \arctan(1/\sqrt{\varphi})$	51.82729°

10. Boundary and Honest Scope

This research makes no claim that $\hat{\pi}$ supersedes the standard transcendental π in any mathematical context where π has been proven transcendentally necessary. The following is stated plainly for scientific integrity:

- The doubling of the cube requires the cube root $\sqrt[3]{2}$, a number of degree three. It is *not* constructible by Wantzel's theorem, and neither the Eye of Horus (powers of $1/2$) nor the Golden Pi triangle (square roots) can reach it. This remains impossible.
- The $abc = 64$ identity and the instrumentum identity are genuine, exact, algebraic identities. They are not *approximations* — they hold to full precision.
- The pyramid connections rely on the standard model (280×440 cubits, Seked 5.5 palms) and on the chosen definition of the cubit as $\varphi^2/5$. They are remarkable coincidences of a constructible constant, not demonstrated physical law.

11. Conclusion

A single algebraic number — the Golden Pi $\hat{\pi} = 4/\sqrt{\varphi}$ — arises from independent sources with a consistency that is difficult to dismiss as noise:

1. It is produced *exactly* by the instrumentum identity $f(x, y) = 4xy^2/((y^2 + x^2)\sqrt{y^2 - x^2})$ at $y/x = \sqrt{\varphi}$.
2. It is encoded in the Great Pyramid through the golden-ratio Royal Cubit $\varphi^2/5$ and the Seked slope $7/5.5 \approx \sqrt{\varphi}$.
3. It completes a right triangle whose sides form the golden progression $\{a, a\sqrt{\varphi}, a\varphi\}$ and whose product is exactly 64 — supplying the missing 1/64 of the Eye of Horus.
4. Its complement angle reproduces the pyramid face angle to within one arcminute.

The golden ratio's square root — simultaneously the pyramid's slope and an algebraic square root — is the unifying spine. Whether one regards this as hidden ancient engineering or as a beautiful coincidence of algebraic numbers, the identities themselves are exact and verifiable. They are offered here as a complete, self-consistent account of everything currently known about the Golden Pi constant.

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